

inside the console understands the orientation of the command-launcher console. The missile has a small IR beacon on its rear whose emissions the command-launcher console continuously tracks. From these two parameters, the command-launcher console understands the missile's trajectory and the target's position. It then calculates the trajectory that the missile should follow. Accordingly, it sends the guidance signals through the uncoiled wire.

The probability of a tank being engaged by any of these two types of missiles is very high. So the SK-APS must be capable of tackling both these types. A typical SK-APS is based on a control computer taking input from laser sensors (situated at strategic locations on the tank's body) and controlling an electro-optical jammer and grenade launchers, which are situated on the sides of the tank.

To tackle laser-guided missiles, the SK-APS computer continuously looks for any laser emissions in the tank's vicinity through its laser sensors. Once the control computer detects a pattern of laser emissions, it activates the grenade launchers present in the direction from where the missile is arriving. Grenade launchers fire smoke grenades that dispense a thick aerosol cloud at a distance from the tank and effectively erect a smoke screen. This cloud is opaque for the lasers from the designators to penetrate. So the laser gets reflected and the missile mistakes the cloud as a target. But unknown to the missile, the tank moves to a different position and escapes from the killer missile.

To defeat wire-guided missiles, the SK-APS, uses a jammer. The jammer is a powerful IR flashlight that can also pulsate. The command-launcher console tracks the IR beacon of the missile to find the position of the missile. When the missile starts approaching the tank, the SK-APS computer activates the electro-optical jammers that start emitting IR pulses. The command-launcher console mistakes the jammer for missile's IR beacon. Since jammer is an integral

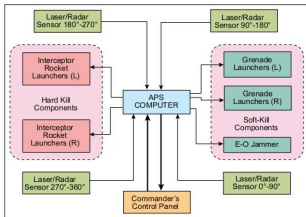


Fig. 7: Block diagram of a typical active protection system showing both hard and soft kill components



Fig. 8: Shtora with jammers active; note the lit red flashlights (Internet photo)

part of the tank, the command-launcher thinks that the missile is exactly on course, but in fact the missile is now off course. The missile therefore crashes somewhere other than the tank.

American AN/VLQ-6 missile countermeasures device, German multi-functional self-protection system and Russian Shtora are some of the predominant SK-APS systems in use.

Hard-kill active protection systems

A major handicap of an SK-APS is its inability to defeat anti-tank rockets. An anti-tank rocket, called rocket propelled grenade (RPG), is a dumb weapon without any electronic guidance system. So it cannot be confused. It just travels in a straight line like a bullet. Though dumb and cheap, it can effectively destroy millions of dollars worth tank. Hence it has become the favourite weapon of terrorists.

The changing geopolitics has led the battles to be fought in cities against terrorists and insurgents rather than

in open terrains. This requires the tanks to prowling the streets, an unfavourable ground for tanks. Even a minimally trained person can take out a tank with an RPG from a roof-top. So narrow streets and gullies have now become tank-killing grounds. This has necessitated hard-kill systems to destroy these rockets. So far only Russia and Israel have had to fight intensely inside cities and faced the wrath of the RPGs. Forced by these conditions, only they have successfully deployed such systems to defeat the RPG menace. Developmental efforts are going on all over the world to develop such systems.

The HK-APS is a fully-automatic system providing a very high degree of protection for the tank in all climatic and terrain conditions. It can protect the tank from anti-tank rockets and missiles, and treats them the same way. It uses radar for detecting the incoming projectile missile or rocket. Then it uses interceptor rocket launchers instead of grenade launchers and jammers.

Multi-directional radar mounted on roof of the tank acts as eyes of the HK-APS and constantly scans for an approaching rocket. Depending on the type and design of the HK-APS, separate radars scanning different sectors may replace this multi-directional radar. The HK-APS computer filters the resultant radar echoes and culls out the echoes matching that of these RPGs.

Once the echo of an RPG is filtered and identified, the radar switches to the target-tracking mode. In this mode, it stops looking elsewhere and continuously looks at the projectile to derive the direction and bearing (trajectory) of the incoming RPG. This trajectory